

PHYSICS-DRIVEN INTERPRETABLE SAR TARGET RECOGNITION NETWORK BASED ON SCATTERING CENTER FEATURE EXTRACTION

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ABSTRACT

Synthetic aperture radar (SAR) target recognition methods based on deep learning have been a research hot-spot. But most of them are of black-box structure and neglect physical characteristics of SAR targets, which restrict the recognition performance. Scattering centers (SCs) are physical features that reflect structure and shape information of targets. Thus, this paper designs a physics-driven interpretable SAR target recognition network based on scattering center feature extraction. Incorporating the scattering center model of SAR target into network, our method is interpretable deep model that learns the SC features with specific physical meanings. Moreover, the learned SC features are then constructed as the SC geometric images, which are further projected into a designed target recognition network for target recognition. In particular, our model is an end-to-end model that achieves SCs feature extraction and target recognition in a framework, which can avoid the mismatch issue between SC features and classifier and ensure promising recognition performance. Experiments on the measured MSTAR dataset validate the superior performance of our method.

Index Terms— Synthetic aperture radar (SAR), target recognition, scattering center model, complex networks, interpretability.

1. INTRODUCTION

Synthetic aperture radar (SAR) is a widely-used technique that can be utilized to acquire lots of high-resolution images in various conditions. SAR target recognition have attracted much attention in remote sensing image processing field.

Traditional SAR target recognition methods are focused on feature extraction. In [1], principal component analysis (PCA) is utilized to extract the subspace features of SAR

images, and then the linear support vector machine (LSVM) is used for target recognition. The K-singularly valuable decomposition (K-SVD) [2] and linear discriminant analysis (LDA) [3] are taken for extracting over-complete features and low-dimensional features, respectively. Ding et al. [4] use the approximate maximum likelihood (AML) algorithm to extract scattering center (SC) features, and then design a template matching method for SAR target recognition.

Recently, since neural networks have powerful feature exploration capability, researchers have already developed various deep SAR target recognition networks. Chen et al. [5] propose the A-ConvNet that is a full convolution network to explore deep features of SAR images by using convolution layers. Ding et al. [6] first utilize backbone structures like VGG16 and VGG19 for SAR target recognition. In [7], a deep methods combining data augmentation techniques is proposed for target robust recognition on SAR images. In [8], based on component analysis, Li et al. develop a multi-scale CNN and have achieved superior results on measured SAR images by fusing local and global features information of SAR images.

Although above methods [5-8] have gained promising results in SAR target recognition, they are data-driven model that neglect physical characteristics of SAR targets and thus are hard to be interpreted. Moreover, most of existing SAR target recognition networks are improved from the backbone model that are proposed for optical images. However, SAR images are very different from optical images with the unique electromagnetic scattering characteristics. Scattering centers (SCs) are physical features of complex SAR images based on imaging theory and contain rich physically relevant descriptions of targets [13].

For the sake of utilizing electromagnetic characteristics of SAR target and deep framework of CNN, we propose a physics-driven interpretable SAR target recognition network based on scattering center feature extraction. Firstly, the scattering center model of SAR target is incorporated into network, and our model is driven by physical mechanism. In this way, our model learns the scattering center (SC) features that reflect target structure and shape size. Then, the SCs are constructed as geometric images, which are then fed into a designed CNN by exploring physical structure information in SC geometric images for target recognition. In particular,

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our method is an end-to-end model that achieves SC features extraction and target recognition in a framework, which achieves promising recognition performance.

We list contributions of our proposed method as follows:

(1) Our method is a physics-driven interpretable model that embeds the scattering center model of SAR targets into network, which is helpful to explore physical features with electromagnetic scattering characteristics.

(2) Our method is an end-to-end model that achieves the integration of scattering center (SC) feature extraction and recognition in a framework, avoiding the mismatch between SC features and classifier.

Next, Section 2 introduces the details of our proposed method. And Section 3 shows some experiments on MSTAR dataset. Finally, the conclusions are summarized in Section 4.

2. PROPOSED METHOD

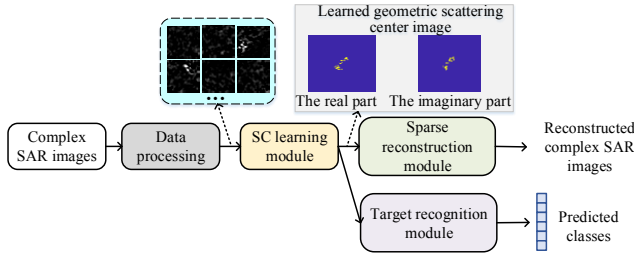


Fig. 1. The overall framework of our method.

Fig. 1 illustrates the overall framework of our method, which consists of SC learning module, the sparse reconstruction module and target recognition module. In our model, the SC learning module aims to learn SC features from the image patches after data processing. Then, the learned SCs are constructed as SC geometric images, which are fed into image reconstruction module that combines the SC model of SAR targets to obtain reconstructed complex SAR images. Simultaneously, the SC geometric images also projected into target recognition module to predict the class of images.

We will successively introduce three modules of our method in detail in the following sections.

2.1. SC learning module

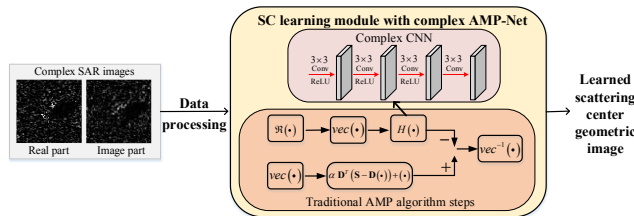


Fig. 2. The network architecture of SC learning module with complex AMP-Net. Convolution operation is represented as “conv”.

According to sparse imaging theory [11], the scattering center model of SAR targets is expressed as:

$$\mathbf{s} = \mathbf{D}\mathbf{z} + \boldsymbol{\varepsilon} \quad (1)$$

where $\mathbf{s}$ and $\mathbf{z}$ respectively denote input SAR image and its corresponding scattering coefficient, $\boldsymbol{\varepsilon}$ denotes the residual

error. As for Eq. (1), scattering center extraction process can be seen as a sparse optimization problem. Thus, a sparse solving algorithm is utilized to extract scattering center based on the physical model of SAR images. Here, we adopt AMP [9] as the sparse solving algorithm. According to the derived steps of AMP algorithm, the scattering coefficient $\mathbf{z}^k$ in the k -th iteration can be derived as:

$$\mathbf{z}^k = \alpha_k \mathbf{D}^T \boldsymbol{\varepsilon}^{k-1} + \mathbf{z}^{k-1} - (\alpha_k \mathbf{D}^T \mathbf{D} - \mathbf{I})(\bar{\mathbf{z}} - \mathbf{z}^{k-1}) \quad (2)$$

with α_k being a trainable parameter and $\bar{\mathbf{z}}$ denoting the expectation of the true value $\mathbf{z}$. In real cases, the prior of $\mathbf{z}$ is needed to be set first, but it is hard to assume the accurate prior distribute of $\mathbf{z}$, and thus the estimation of $\bar{\mathbf{z}}$ is not accurate. Recently, deep unfolding networks [9] have been proposed for tasks of compressive image sensing and anomaly detection. Inspired by these methods, we design a SC learning module with complex AMP-Net based on the SC model. After data processing step, input image is divided into m patches as $\sigma(\mathbf{s}, m)$, and its corresponding scattering coefficient $\mathbf{z}$ is divided into m patches as $\sigma(\mathbf{z}, m)$. Thus, the scattering coefficient $\mathbf{z}^k$ in the k -th iteration can be derived as:

$$\mathbf{z}^k = \sigma^{-1} \left(\text{vec}^{-1} \begin{pmatrix} \alpha_k \mathbf{D}^T R^{k-1} + \text{vec}(\sigma(\mathbf{z}^{k-1}, m)) \\ -H^k \text{vec}(\Re_k(\sigma(\mathbf{z}^{k-1}, m))) \end{pmatrix}, m \right) \quad (3)$$

$$H^k = \alpha_k \mathbf{D}^T \mathbf{D} - \mathbf{I} \quad (4)$$

$$R^{k-1} = \mathbf{s} - \mathbf{D} \text{vec}(\sigma(\mathbf{z}^{k-1}, m)) \quad (5)$$

where $\text{vec}(\cdot)$ denotes the vectorization operation, and $\Re_k(\cdot)$ is introduced to fit the term $(\bar{\mathbf{z}} - \mathbf{z}^{k-1})$ that is achieved by the convolution neural networks (CNNs) with multiple layers. Since the scattering coefficient $\mathbf{z}^k$ is complex-valued, CNNs are constructed in a complex-valued neural network framework. For clear illustration, Fig. 2 shows the detailed architecture of SC learning module with complex AMP-Net.

2.2. Image reconstruction module

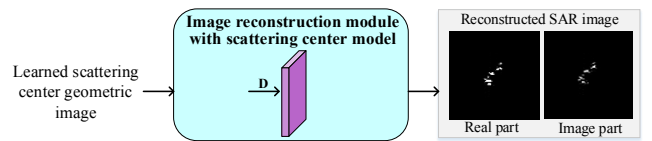


Fig. 3. The network architecture of sparse reconstruction module.

The network architecture of sparse reconstruction module in our model is shown in Fig. 3. To incorporate the SC model into network, a single liner full connection layer, whose the weight matrix is fixed as Fourier dictionary $\mathbf{D}$ in image domain, is designed, as illustrated in Fig. 3. Given scattering coefficient $\mathbf{z}$, reconstructed image $\hat{\mathbf{s}}$ can be generated by:

$$\text{Latent layer: } \hat{\mathbf{s}} = \mathbf{D}\hat{\mathbf{z}} \quad (6)$$

where $\mathbf{D}$ denotes the weight matrix of a full connection layer, and $\hat{\mathbf{z}}$ represents the estimated scattering coefficient $\mathbf{z}$. In particular, during the image reconstruction process,

the input, output and network parameter are complex-valued. Since the input are sparse scattering coefficients, the output image can remove the clutters and noises in the background area, and simultaneously remain shape and pixels of targets.

Thus, our model is a physics-driven model that embeds the physical model of SAR images into networks. The output of this module is SCs comprising of the locations and amplitude of dominant scattering centers in SAR targets, which reflect important physical characteristics about target structure.

2.3. Target recognition module

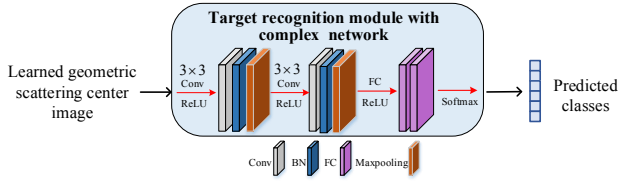


Fig. 4. The network architecture of target recognition module with complex network.

In Fig. 4, the network architecture of target recognition module is consisted of complex network with multiple layers. After SC learning module, the SCs containing the locations and amplitude for an input image are constructed as an SC geometric image, which is fed into target recognition module to obtain class prediction. In detail, SC geometric image contains the physical structure information, such as the size, shape and electromagnetic scattering characteristics and so on. In addition, CNNs is capable of exploring the features of points, lines and outlines in images. Therefore, we design a target recognition network with CNNs to explore the useful information from SC geometric image for recognition.

3. EXPERIMENTAL RESULTS AND ANALYSIS

This section utilizes MSTAR dataset to conduct experiments, and the dataset is consisted of measured SAR images with 0.3×0.3 m resolution that describes military ground targets. We adopts the three-target data to conduct experiments on our method, and the details of the data are listed in TABLE 1.

TABLE 1. THE TARGET INFORMATION OF THREE-TARGET DATA

Dataset	BMP2			BTR70		T72	
	C21	9566	9563	C71	132	S7	812
Training (17°)	233	0	0	233	232	0	0
Test (15°)	196	196	195	196	196	191	195

In the following, we first show ablation study results of our method. Then, we compare the recognition performance of our method with related methods, and also validate the noise robust performance as well.

3.1. Ablation study

For our method, we design some controlled experiments to verify the respective effect of scattering center extraction and end-to-end framework on recognition performance.

TABLE 2 lists the recognition results on the measured three-target MSTAR data. In TABLE 2, the last row is our method, and the probability of correct classification (PCC) is calculated amount all number of test samples. Compared with the result in first row, our method achieves much higher recognition performance by learning SCs that contain abundant target structure information. The results in second and third rows illustrate that the SCs obtained by our method a little outperform those obtained by OMP algorithm [10]. Moreover, the results in third and fourth rows shows that our end-to-end method achieves much better performance. Therefore, our method can extract the SCs that are beneficial to recognition, and the end-to-end framework is effective to improve performance.

TABLE 2. THE EXPERIMENTAL RESULTS OF ABLATION STUDY ON THREE-TARGET MSTAR DATA

Components?				PCC
Using SAR images	Extracting SCs by OMP	Extracting SCs by our method	End-to-end	
✓	×	×	✓	0.8491
×	✓	×	×	0.8901
×	×	✓	×	0.8915
×	×	✓	✓	0.9815

3.2. Comparisons with other methods

In this sub-section, we first compare our method with some related methods, including original SAR images, PCA [1], template matching [4], DL-JDSR [12], AE, DAE, Euclidean-AE, VGGNet [6] and A-ConvNet [5]. The recognition results on the three-target data are illustrated in Fig. 5, which shows that our method obtains much higher PCC than other methods by exploring SC features that contain physical structure of SAR targets.

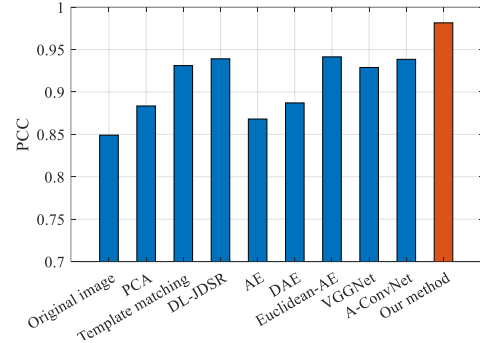


Fig. 5. Results of noised SAR images under different SNRs.

Moreover, to validate the noise robust performance of our method, some experiments are conducted under different

SNRs. In Fig. 6, the first row lists noised test images under -10dB, -5dB and 0dB, and the second row are the learned SC features by our methods, which shows that the SC features well remove noise and retain target. The reconstructed SAR images in the third row also perform well under low SNRs. Thus, Fig. 6 illustrates that our method can learn the SCs of SAR targets that are noise robust. Moreover, to compare the noise robust recognition performance of our method with other methods, Fig. 7 presents the results under different SNRs. As shown in Fig. 7, our method always gains much higher results than other methods under all SNRs, which validate promising noise robust recognition performance of our method by learning robust SC features.

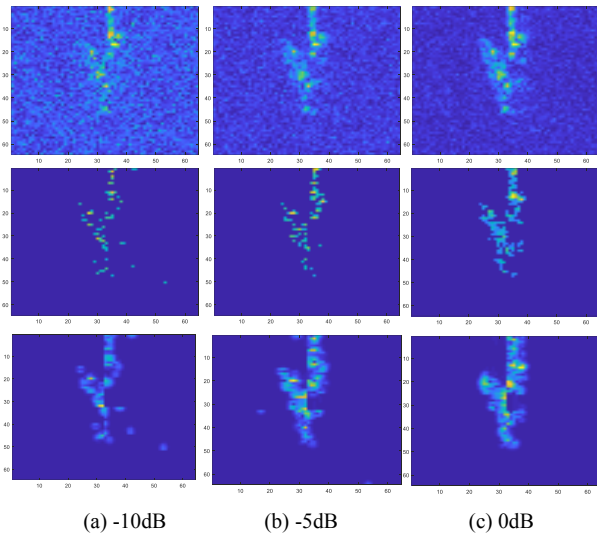


Fig. 6. Results of noised SAR images under different SNRs. The first row shows the original SAR images, and the second row shows the SC geometric images, and the third row shows the reconstructed images.

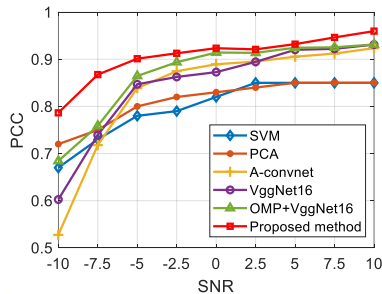


Fig. 7. Results of target recognition methods under different SNRs.

4. CONCLUSION

Due to the defects of black-box structure for most existing SAR target recognition networks, this paper develop a novel physical interpretable method by incorporating SC model of SAR target into network. Our method learns the SC features with specific physical meanings, and thus they are easy to be interpreted. Moreover, the SC features are then constructed as the geometric images, which are projected into a designed complex CNN for target recognition. Our model achieves

the SC feature extraction and target recognition in an end-to-end framework, which can help learn the features that are useful for recognition. Experiments on measured MSTAR dataset show that our method gains much better recognition performance than other methods, and also shows noise robust performance under low SNRs.

5. REFERENCES

- [1] Q. Zhao and J. C. Principe, "Support vector machines for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 643–654, Apr. 2001.
- [2] M. Aharon, M. Elad, A. Bruckstein, "K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Transactions on Signal Process*, vol. 54, no. 11, pp. 4311–4322, 2006.
- [3] C. M. Bishop. "Pattern Recognition and Machine Learning," Springer, pp. 559-599, 2006.
- [4] B. Ding, G. Wen, J. Zhong, C. Ma, and X. Yang, "A robust similarity measure for attributed scattering center sets with application to SAR ATR," *Neurocomputing*, vol. 219, pp. 130–143, Jan. 2017.
- [5] S. Chen, H. Wang, F. Xu, and Y. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.
- [6] J. Ding, B. Chen, H. Liu, and M. Huang, "Convolutional neural network with data augmentation for SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 364–368, Mar. 2016.
- [7] L. E. Falqueto, J. A. S. Sa, R. L. Paes, and A. Passaro, "Oil rig recognition using convolutional neural network on Sentinel-1 SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1329–1333, Aug. 2019.
- [8] Y. Li, L. Du and D. Wei, "Multiscale CNN based on component analysis for SAR ATR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1-12, 2022.
- [9] Z. Zhang, Y. Liu, J. Liu, F. Wen and C. Zhu, "AMP-net: Denoising-based deep unfolding for compressive image sensing," *IEEE Trans. Image Process.*, vol. 30, pp. 1487–1500, 2021.
- [10] J. A. Tropp and A. C. Gilbert, "Signal recovery from random measurements via orthogonal matching pursuit," *IEEE Transactions on information theory*, vol. 53, no. 12, pp. 4655–4666, 2007.
- [11] L. Zhang, M. Xing, C.-W. Qiu, J. Li, J. Sheng, Y. Li, and Z. Bao, "Resolution enhancement for inversed synthetic aperture radar imaging under low snr via improved compressive sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3824–3838, 2010.
- [12] Y. Sun, L. Du, Y. Wang, Y. Wang, and J. Hu, "SAR Automatic Target Recognition Based on Dictionary Learning and Joint Dynamic Sparse Representation," *IEEE Geoscience & Remote Sensing Letters*, vol. 13, pp. 1777-1781, 2017.
- [13] Y. Li and L. Du, "Design of the physically interpretable SAR target recognition network combined with electromagnetic scattering characteristics", *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, pp. 4988-4991, Jul. 2022.